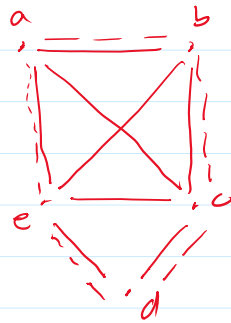


lecture 21.

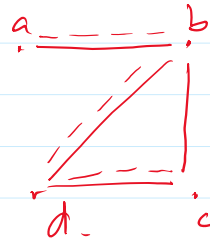
HAMILTON PATH :- Traverses all Vertices.

" Circuit. " " " + Start & end.

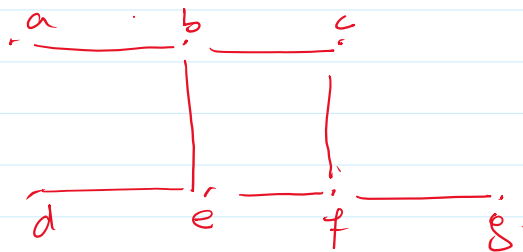
12x5
577



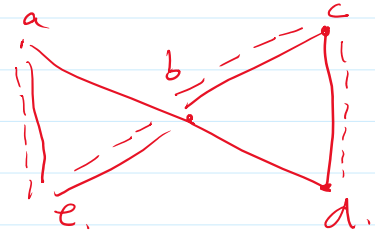
HP ✓
HC ✓



HP ✓
HC X.

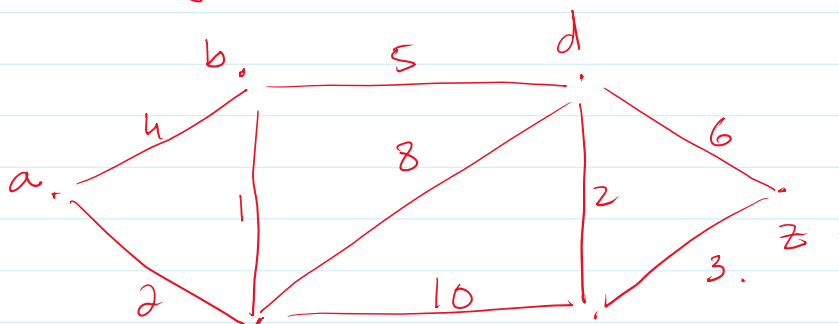


HP X
HC X.



HP
HC.

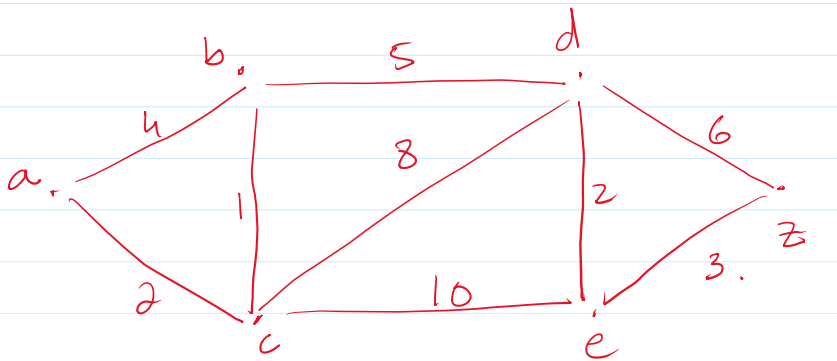
Shortest Path -
Dijkstra.



a to z.

	a	b	c	d	e	z
a	∞	∞	∞	∞	∞	∞
c	∞	$4a$	$2a$	∞	∞	∞
b	∞	$3c$	$2a$	$8b$	$12c$	∞
d	∞	$3c$	$2a$	$8b$	$10d$	$14d$
e	∞	$3c$	$2a$	$8b$	$10d$	$13e$
z	u	u	u	u	u	u

Path . Cost .
acbedez . 13

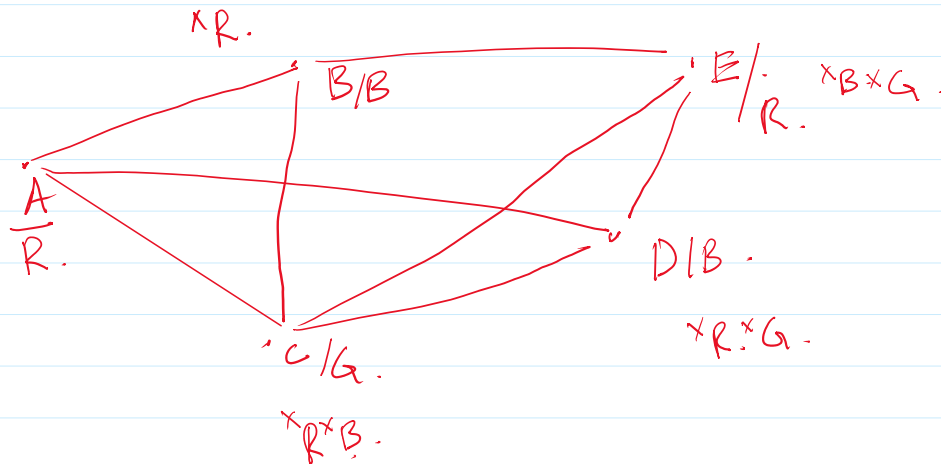
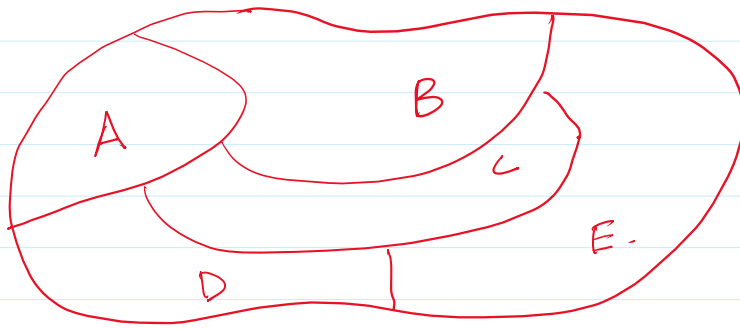


d to a.

	a	b	c	d	e	z
a	∞	∞	∞	∞	∞	∞
d	∞	$5d$	$8d$	0	$2d$	$6d$
e	∞	$5d$	$8d$	0	$2d$	$5e$
z	∞	$5d$	$8d$	0	$2d$	$5e$
b	$9b$	$5d$	$6b$	0	$2d$	$5e$
c	$8c$	$5d$	$6b$	0	$2d$	$5e$
a	u	u	u	u	u	u

Path . Cost .
d.bca . 8

Graph Colouring .



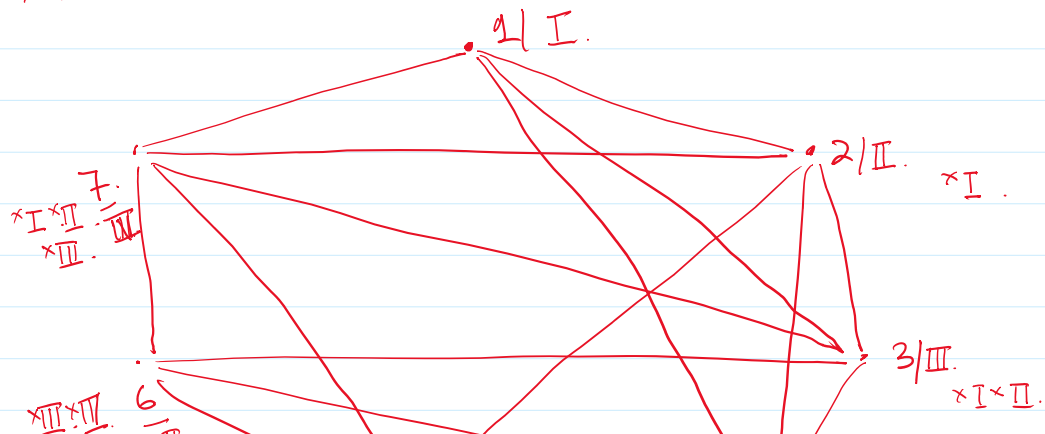
R, B, G, O, W, I, ...

Chromatic Number 23.

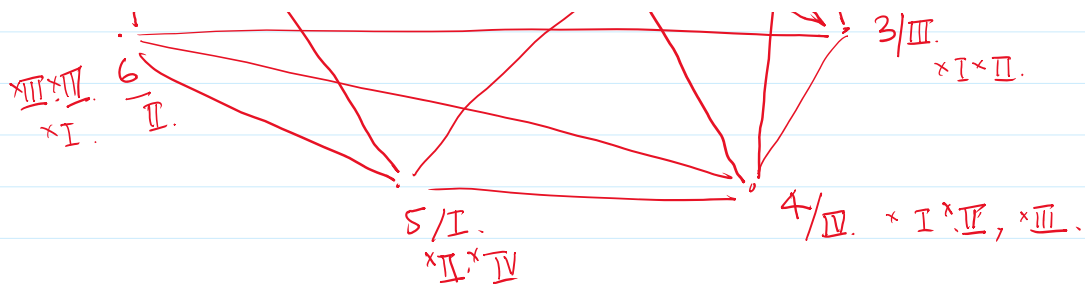
Chromatic Number: Minimum Number of Colours to Color a graph such that no two adjacent Vertices has the same Colour.

Ex 5	Courses.	1 & 2	2 & 3	3 & 4	4 & 5		
6 II.		1 & 3	2 & 4	4 & 6	4 & 6		
		1 & 4	2 & 5	4 & 7			
		2 & 7	2 & 7		5 & 6	6 & 7	
					5 & 7		

I, II, III, IV, ...

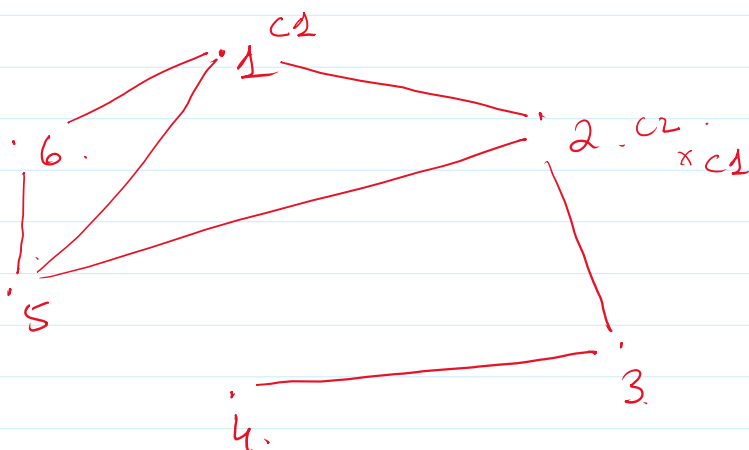


I: 1, 5
II: 2, 6
III: 3
IV: 4, 7



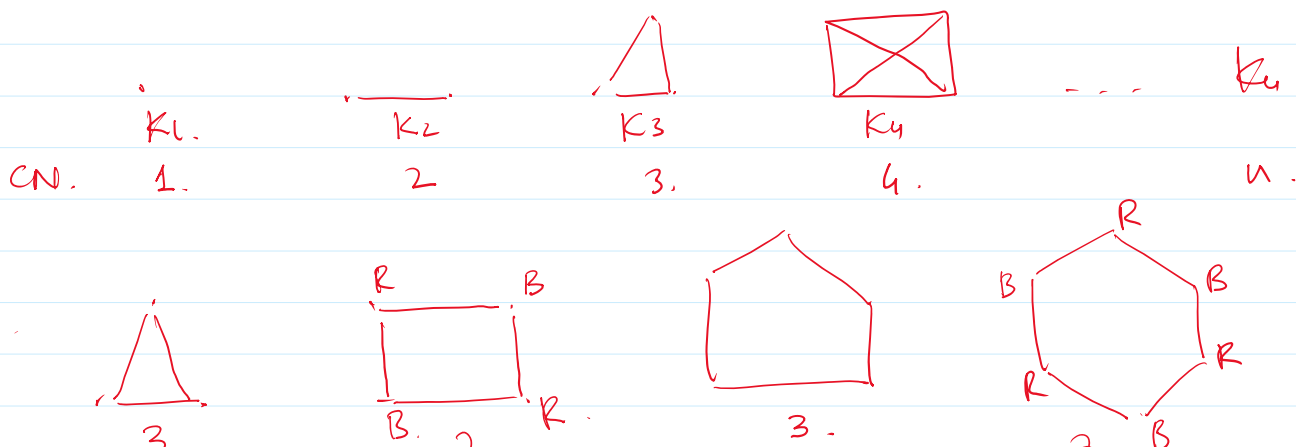
Ex 18
613.

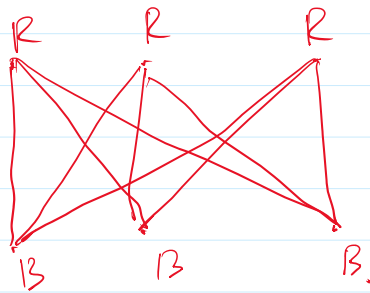
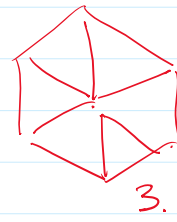
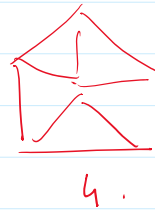
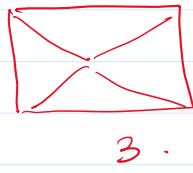
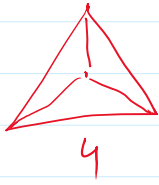
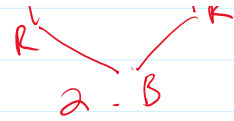
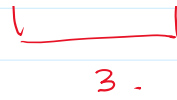
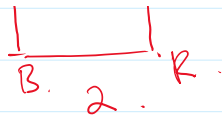
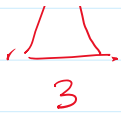
	1	2	3	4	5	6
1	—	85	175	200	50	100
2		—	125	175	100	160
3			—	100	200	250
4				—	210	220
5					—	100
6						—



Resource =
Constraint =

Colours =
Edges =





$K_{3,3}$

